WELLESLEY COLLEGE • DATA SCIENCE MAJOR CAPSTONE • Spring 2026

Mapping Inequality on the Boston Plate

Neighbourhood Socioeconomics & Food Inspection Outcomes in Boston, 2015–2025

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RESEARCH QUESTION

Do food safety inspection outcomes in Boston systematically differ across neighbourhoods of varying socioeconomic status — and has any such gap widened over the past decade?

Yes, but the effect is small — and it has not widened. Establishments in low-income tracts have a ~3.4 percentage-point lower predicted pass probability than those in high-income tracts, and this gap has remained statistically stable across the 2015–2025 window. Most variation in pass/fail is within-establishment noise rather than neighbourhood-level disparity.

i. Background & Motivation

- Food inspection records document **day-to-day enforcement** of health regulations across an entire city.
- If neighbourhood disadvantage shapes pass/fail outcomes, it would suggest **disparities in regulatory protection** across communities.
- Boston offers a **10-year, geocoded record** linked to census tract demographics.
- This creates a natural test case: does SES predict outcomes *after* accounting for establishment-level differences?

ii. Data Sources

Ten years of Boston inspections linked to tract-level Census demographics.

- **Boston Food Establishment Inspections** (2015–2025) — 872,111 raw violation records, deduplicated to 100,211 unique establishment-visit inspections.
- **ACS 2019 5-year estimates** at census-tract level: income, poverty, racial composition, foreign-born share.

→ Spatially joined to inspections via tract polygons.

Why is the fail rate 46%?

- Boston uses an **inclusive definition**: any inspection with at least one violation of any severity.
- Most failures are **minor, correctable issues** — not public health emergencies.

91.6K	6,912	174	46%
INSPECTIONS	ESTABLISHMENTS	CENSUS TRACTS	OVERALL FAIL RATE

iii. Methods

- **Mixed-effects logistic regression** via `glmmTMB` with random intercepts by establishment.
- **Sequential model building**: null → income-only → SES composite → fully adjusted → income × time interaction.
- **PCA-based SES index** to address collinearity ($r = -0.81$ income–poverty). PC1 captures 79%.
- **Train/test split** by establishment (80/20) to avoid leakage.
- **Robustness**: COVID exclusion, nonwhite check, DHARMa diagnostics.
- **Model diagnostics**: residual analysis via DHARMa, AUC evaluation on held-out establishments.

★ KEY FINDING

A 1 SD increase in tract income (~$40K) raises the odds of passing by 6.4%. A low-income Food Service establishment has a 50.1% predicted pass probability, vs 53.5% in a high-income tract. The gap is real but small.

iv. Exploratory Findings

Tract-level fail rate vs median income: **weakly negative**, visually flat. Implication is that any neighbourhood SES effect will be **modest**.

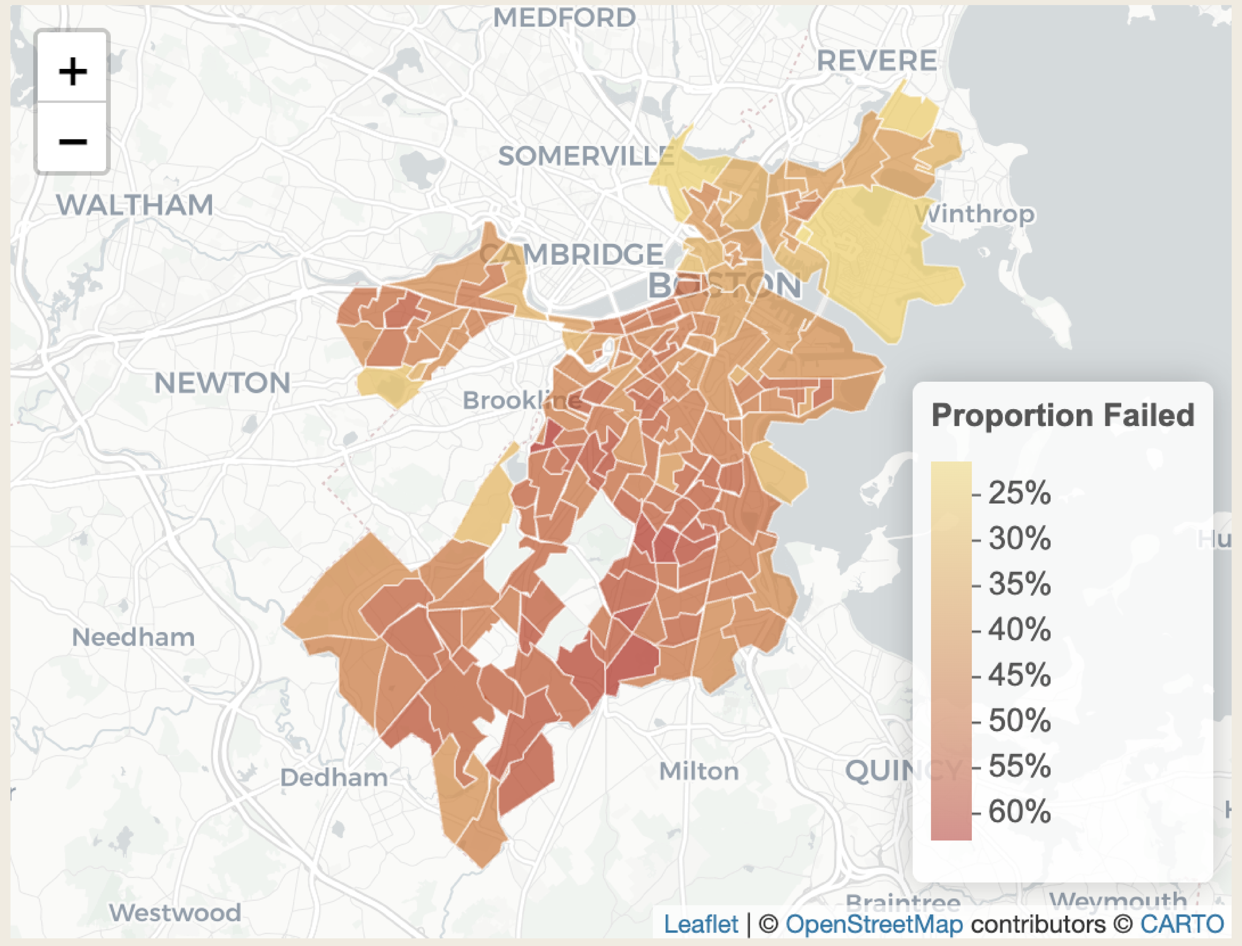

Figure 1. *Inspection fail rates by Boston census tract. Darker tracts indicate higher proportions of failed inspections; the spatial pattern motivates the neighbourhood-level analysis.*

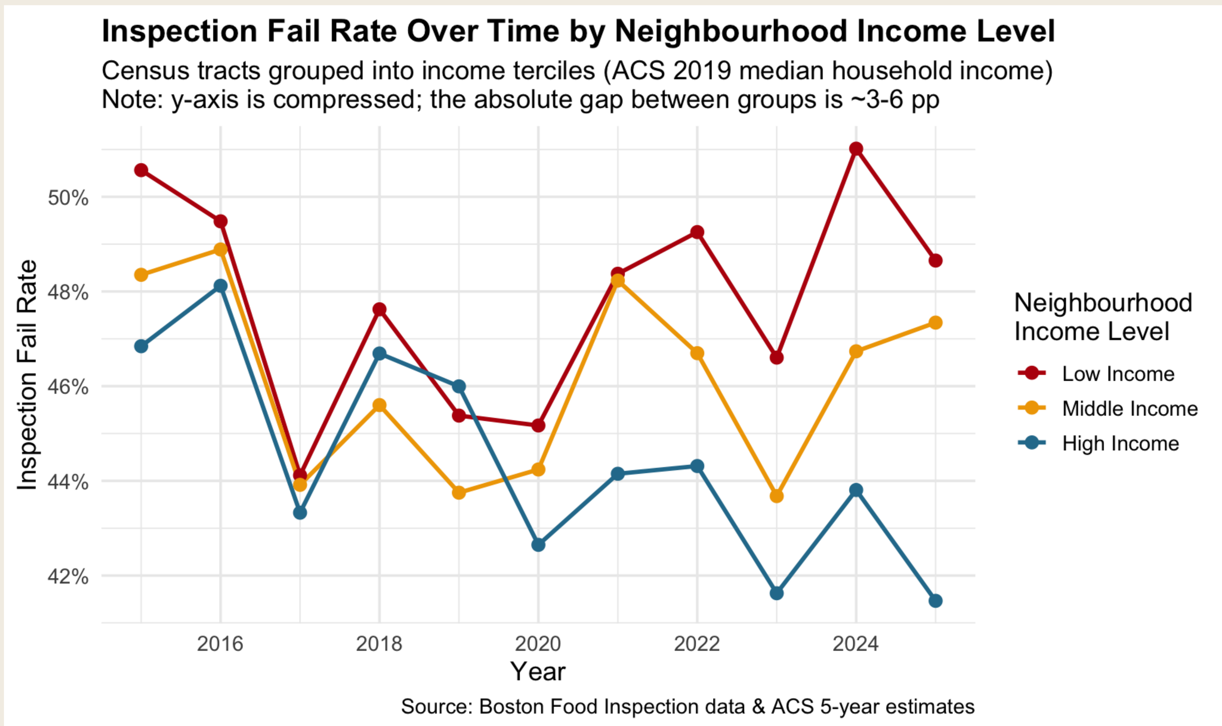

Figure 2. *Tract-level fail rates show only a weak negative slope across the income distribution. Each bubble is a census tract sized by inspection volume.*

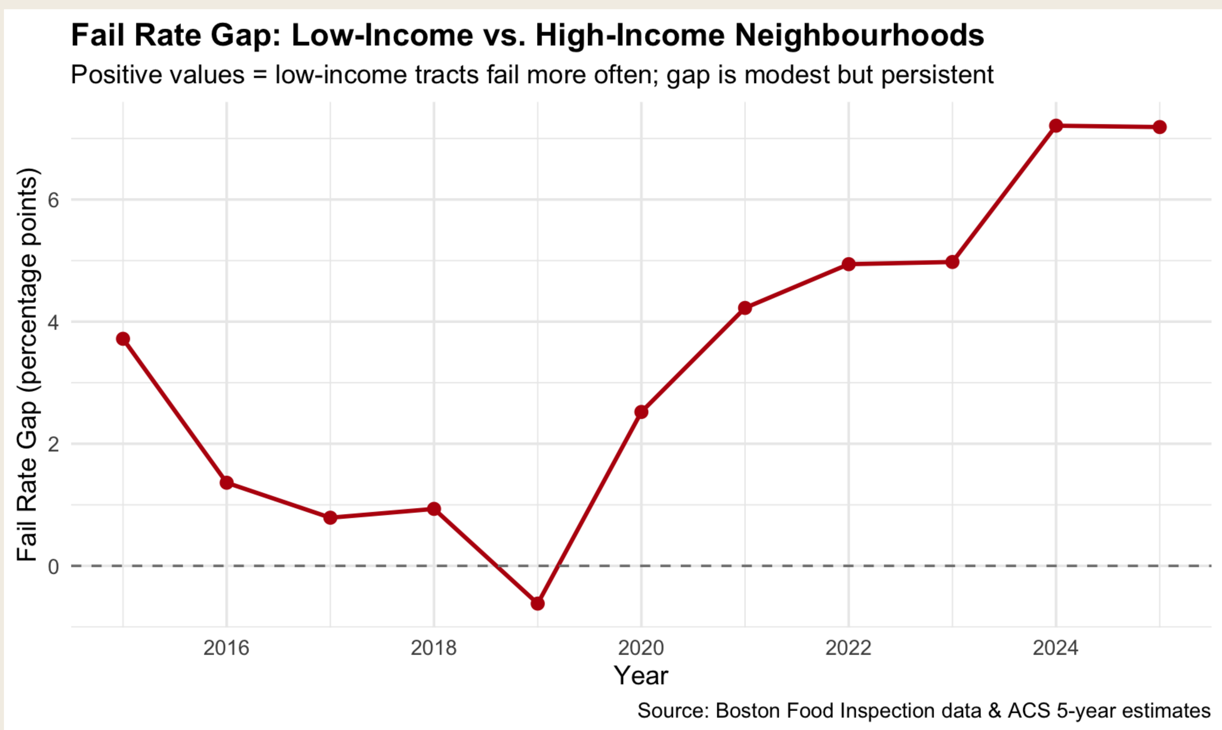

Figure 3. *Low-income tracts consistently fail more often than high-income ones, but the absolute gap is only ~3–6 pp on a compressed y-axis.*

v. Modelling Results

- **Null model ICC = 0.046** — only 4.6% of pass/fail variation lies *between* establishments.
- The remaining **95% is within-establishment** noise (inspector, day, conditions).
- This places a **hard ceiling** on how much any neighbourhood-level predictor can explain.
- Adding fixed effects shrinks the random-effect variance by **87.7%** → little residual establishment heterogeneity.

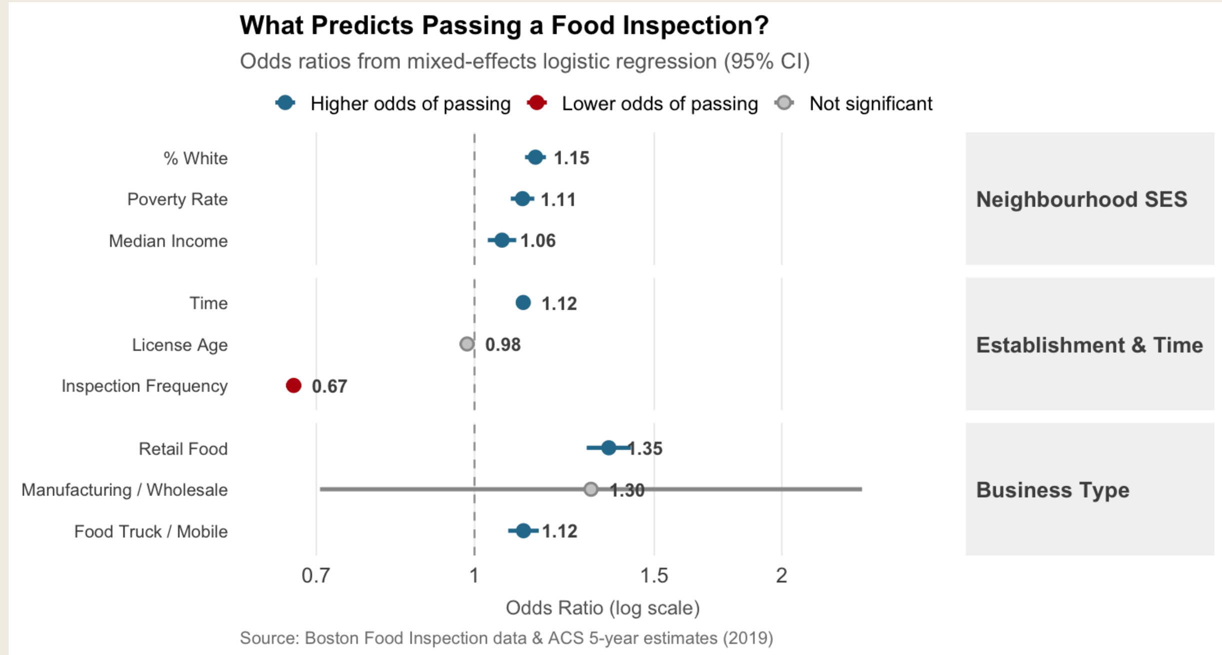

Figure 4. *Inspection frequency dominates the coefficient plot (OR ≈ 0.67). Median income is positive but modest (OR ≈ 1.06).*

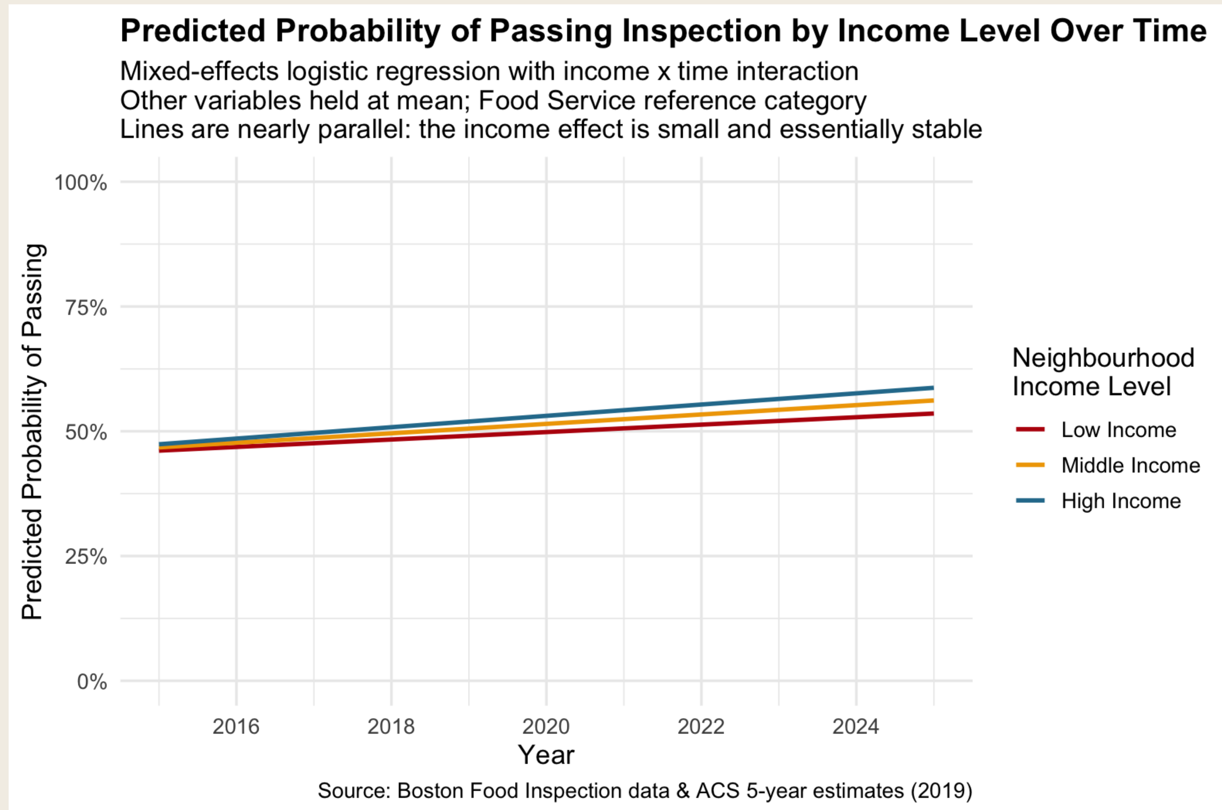

Figure 5. *Predicted-probability lines for low/middle/high income tracts are nearly parallel — the income × time interaction is significant ($p = 0.006$) but practically negligible (OR = 1.022).*

Limitations

- Single-city analysis.
- ACS 2019 demographics applied to a 10-year window.
- Pass/fail is coarse — doesn't capture violation severity.
- Inspection frequency is endogenous to outcomes.
- Spatial autocorrelation not modelled.
- COVID sensitivity check: all ORs shift < 0.02; AUC $= 0.624$.

vi. Interpretation Caveats

- **Poverty coefficient flips sign** when conditioning on income and % White (OR 0.98 → 1.11).
  - Classic *suppression effect* from collinearity.
  - *Not* evidence that poverty improves outcomes.
  - PCA SES index (OR = 1.065) is the cleanest single measure.
- **Inspection frequency is endogenous**:
  - Failures trigger re-inspections.
  - Conditioning on frequency risks *collider bias* on the SES coefficient.
  - Most consequential model limitation — future work should refit without it.
- **Business type explains more than SES**:
  - Mobile food: OR ≈ 1.42.
  - Retail food: OR ≈ 1.18.
  - *What* an establishment does matters more than *where* it sits.

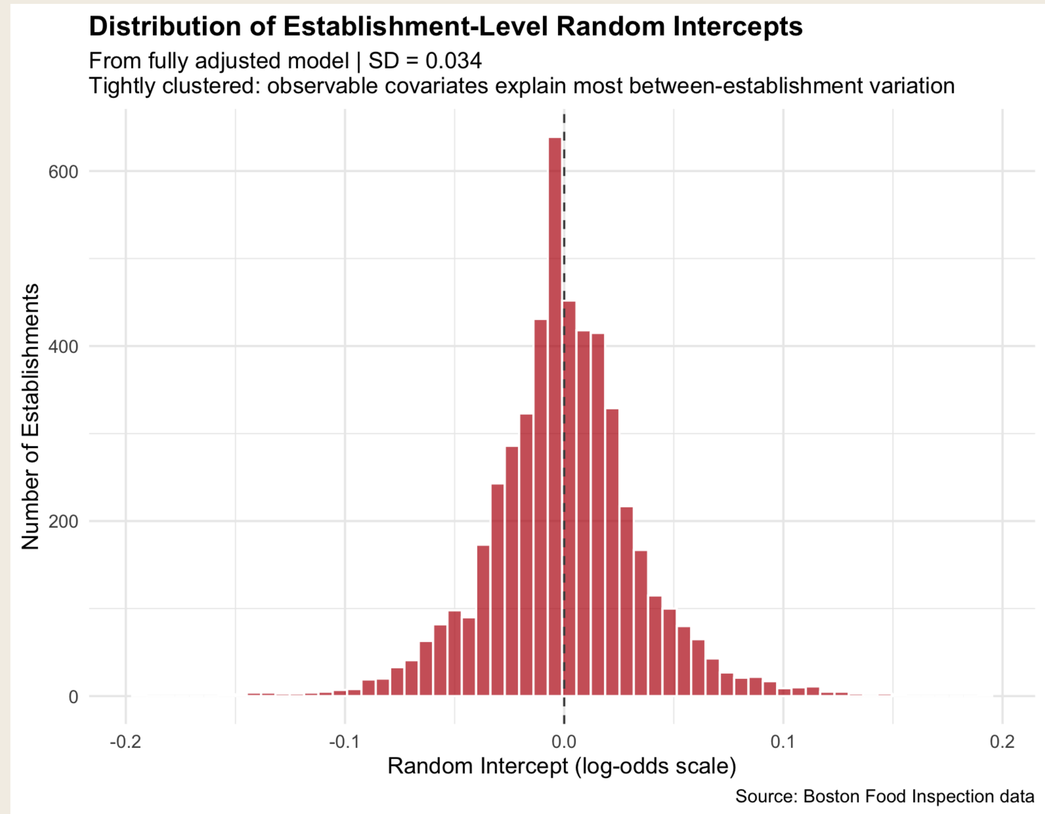

Figure 6. *Establishment random intercepts cluster tightly around zero (SD = 0.034) once observable covariates are accounted for — confirming little residual heterogeneity.*

vii. Conclusion

Neighbourhood income has a real but modest association with food inspection outcomes in Boston.

- Effect is **dwarfed** by within-establishment variability.
- Differences across **business types** matter more than neighbourhood SES.
- Income gradient is **statistically stable** over the past decade.

Policy implication:

- Supports **establishment-level** rather than area-based regulatory attention.
- Pushes back against narratives that low-income Boston neighbourhoods fail inspections at dramatically higher rates.
- *Caveat:* whether establishment-level intervention would actually *improve* outcomes is a separate question this analysis cannot answer.